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Integrative single-cell and spatial transcriptomic analysis reveals a lactate-driven crosstalk between NFATc4⁺ tumor cells and SPP1⁺ macrophages in glioblastoma

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Abstract

Background Glioblastoma (GBM) remains a lethal brain tumor with limited therapeutic options. Metabolic reprogramming, particularly lactate metabolism, plays a critical role in tumor progression and immune evasion.

Methods Here, we integrated single-cell RNA sequencing (scRNA-seq), spatial transcriptomics, and machine learning to investigate the heterogeneity of lactate metabolism in GBM.

Results Using scRNA-seq data (GSE138794), we classified neoplastic cells into high- and low-lactate subgroups and identified nuclear factor of activated T-cells cytoplasmic 4 (NFATc4) as a key transcription factor associated with elevated lactate metabolism. Pseudotime trajectory analysis revealed dynamic upregulation of NFATc4 during neoplastic cell differentiation, correlating with invasive phenotypes. Spatial transcriptomics (GSE194329) demonstrated colocalization of NFATc4⁺ tumor cells with SPP1⁺ macrophages, suggesting their microenvironmental crosstalk. A prognostic model constructed via 101 machine-learning algorithms (StepCox[forward] + RSF) achieved favorable performance across independent cohorts (TCGA, CGGA, GSE108474, GSE4412, and meta-cohort). Patients with different risk levels showed distinct immune infiltration, copy number alterations, and drug-sensitivity profiles. In vitro functional assays confirmed that NFATc4 knockdown in GBM cells suppressed tumor cell proliferation, migration, and cell cycle while promoting apoptosis. Besides, NFATc4 knockdown in GBM cells reduced SPP1 expression, migration, and M2-like polarization of co-cultured macrophages. Moreover, silencing SPP1 in macrophages attenuated the pro-tumorigenic effects on co-cultured GBM cells, validating the functional relevance of the NFATc4–SPP1 axis.

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Conclusion Our study reveals lactate metabolism-related NFATc4 as a promising therapeutic target in GBM, with implications for prognosis stratification and combination therapy.

Keywords Glioblastoma, NFATc4, Lactate metabolism, Single-cell sequencing, Spatial transcriptomic

Introduction

Glioma, originating from neural stem or progenitor cells, is the most common and highly lethal primary tumor of the central nervous system [1]. According to the World Health Organization (WHO) classification, gliomas are divided into four grades, which better define their malignancy and clinical risk. Among these, glioblastoma (GBM, IDH-wildtype, CNS WHO grade 4) is the most aggressive and malignant subtype [2]. Characterized by rapid cellular proliferation, extensive angiogenesis, and inevitable recurrence, GBM confers a dismal prognosis with a 5-year survival rate of 5% under standard care [3]. With the advent of the precision medicine era, gene-targeted therapies have opened new avenues for treating GBM. To date, numerous therapeutic targets have been identified for GBM treatment. Isocitrate dehydrogenase (IDH) enzymes catalyze the oxidative decarboxylation of isocitrate, which is regarded as a critical molecular hallmark of malignant gliomas [4]. O6-methylguanine-DNA methyltransferase (MGMT) is a DNA repair protein associated with patient survival. Additional glioma-associated biomarkers, such as 1p/19q co-deletion, EGFR, CD133, and alterations in the p53 signaling pathway, have been successively discovered [5]. These discoveries have significant implications for GBM treatment, driving advances in novel therapeutic strategies, including immunotherapy, targeted therapy, and gene therapy.

Over the past few decades, immunotherapy has proven effective in cancer therapy. Increasing immune sensitivity and cytotoxicity can effectively restrain tumor growth. However, its efficacy is often challenged by the complexity of the tumor microenvironment (TME) [6]. TME is mainly composed of immune cells, tumor cells, and other stromal cells. These cells interact with and influence each other within the tumor microenvironment. In the TME, several tumor-associated immune cell populations are present, including immunosuppressive tumor-associated macrophages (TAMs) and regulatory T cells (Tregs). Crucially, these cellular interactions are deeply influenced by the tumor's metabolic landscape. Understanding the mechanism of these cells may promote the efficacy of immunotherapy. Cellular metabolism provides energy for cancer cells to grow and metastasize. Compared to normal cells, cancer cells exhibit a distinct metabolic pattern that enables them to sustain their high energy demand [7]. More importantly, this metabolic dysregulation actively remodels the TME, contributing to the immunosuppressive phenotype. Therefore, targeting these specific metabolic patterns offers a dual benefit: suppressing tumor growth and alleviating the immunosuppressive barrier, providing a promising therapeutic strategy.

Lactate metabolism is one of the most abundant metabolic processes in the TME. In glycolysis, this process converts pyruvate into lactate, a well-known metabolic waste product. Lactate metabolism plays a crucial role in tumor cells, differing from that in normal tissue, a phenomenon first described by Otto Warburg in the 20th century. In recent years, increasing evidence suggests that lactic acid may create a favorable environment for cancer cells by impairing T cell function. For example, previous studies have shown that G-protein-coupled receptor 81 (GPR81) may be activated by lactic acid and promote tumor growth [8]. In gastric cancer models, lactic acid induces CX3CL expression and promotes Treg cell proliferation, thereby facilitating tumor cell growth [9]. Besides, lactic acid can also facilitate the degradation of cyclic GMP-AMP synthase (cGAS) in lung adenocarcinoma [10]. However, in GBM, the influence of lactate metabolism on the immune system remains unclear. Therefore, exploring the role and mechanisms underlying lactate metabolism in glioma represents a promising approach to developing targeted therapies.

Crucially, the metabolic reprogramming of tumor cells is often driven by specific transcriptional regulatory networks that coordinate gene expression in response to TME cues. Nuclear factor of activated T-cells, cytoplasmic 4 (NFATc4) is a member of the NFAT family, and the Ca2+/calcineurin signaling activates it. It is now clear that NFAT proteins play a crucial role in the development and function of the immune system, particularly in T cell-associated functions [11]. The biological function of NFATc4 is mediated by the SP motifs' phosphorylation and dephosphorylation. Due to its low expression in tissue, previous studies have often overlooked the function of NFATc4 in cancer [12]. NFATc4 has been shown to play a key role in various diseases, including cardiovascular and other diseases [13, 14]. With advances in detection and research technologies, an increasing number of studies demonstrate a link between NFATc4 and tumor progression. In lung adenocarcinoma, NFATc4 promotes tumor proliferation by activating the CCNB1/CDK1 pathways [15]. And in ovarian cancer, studies have shown that NFATc4 acts as a quiescence factor [16]. Also in AML, NFATc4 recruits Tregs to mediate immune suppression [17]. Collectively, these findings suggest that NFATc4 plays an important role in tumor progression.

However, its underlying mechanisms in GBM remain unclear.

Notably, while focusing on individual transcription factors such as NFATc4 provides important mechanistic insights into lactate-driven tumor-immune interactions, tumor progression is ultimately governed by complex metabolic reprogramming at the system level. The broader impact of dysregulated lactate metabolism on patient prognosis and therapeutic response remains difficult to predict. Thus, there is an urgent need to translate these complex metabolic features into robust and clinically applicable predictive models.

In this study, we employed single-cell RNA sequencing (scRNA-seq) profiles to investigate the differences in lactate metabolism among neoplastic cells in GBM samples. We performed pySCENIC analysis to identify the key molecular targets associated with high and low levels of lactate metabolism. Furthermore, we used spatial transcriptomic analysis to identify cell populations that are spatially correlated with high-lactate neoplastic cells. Meanwhile, a prognostic model for GBM was developed using 101 machine learning algorithms based on a gene set related to lactate metabolism, and its predictive performance was validated across multiple independent datasets. We also performed relevant in vitro assays for validation. Finally, we found that in the high-lactate GBM, NFATc4⁺ neoplastic cells and SPP1⁺ macrophages synergistically promote tumor growth. The overall workflow and mechanistic framework of this study are shown in the schematic diagram (Fig. 1).

Materials and methods

Collection and integration of the data

Transcriptomic profiles and clinical annotations were available from publicly accessible repositories. The Bulk RNA-seq data for GBM samples include The Cancer Genome Atlas (TCGA, expression values transformed to TPM), the Chinese Glioma Genome Atlas (CGGA, http://www.cgga.org.cn/), and the Gene Expression Omnibus (GEO, https://www.ncbi.nlm.nih.gov/geo/) datasets with accession IDs GSE108474 and GSE4412.

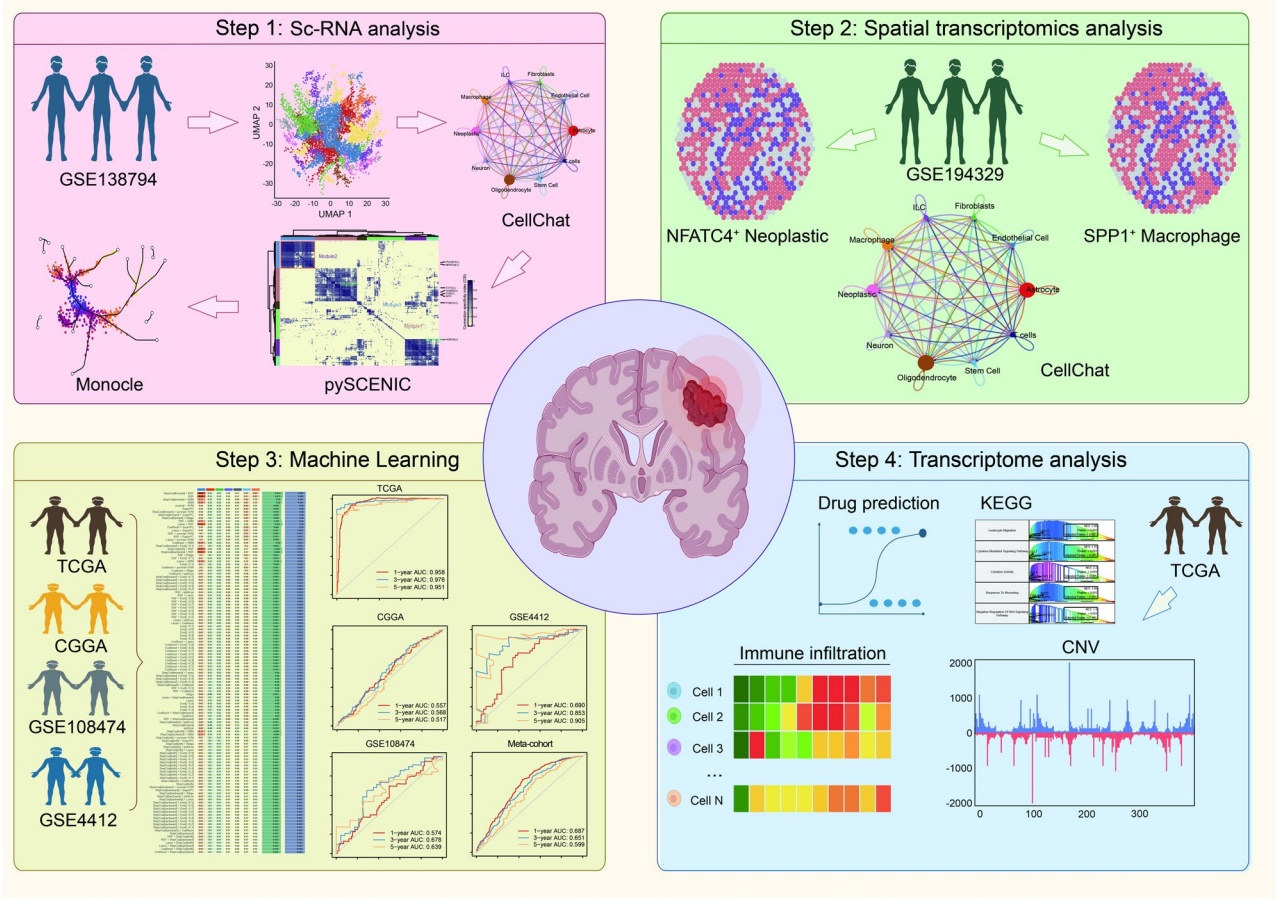

Fig. 1 Schematic overview of the integrative analysis pipeline. The workflow begins with scRNA-seq analysis of GBM samples (GSE138794) to identify transcription factors associated with lactate metabolism. NFATc4 is selected as the key regulator. Spatial transcriptomics (GSE194329) is then employed to investigate cellular interactions and spatial distribution. Multi-dataset validation, including machine learning, immune infiltration profiling, and copy number variation analysis, confirms the prognostic and mechanistic relevance of NFATc4 in GBM

The scRNA-seq data (GSE138794), comprising five GBM datasets, were selected for analysis.

Spatial Transcriptomic sequencing data (GSE194329) were also integrated into the investigation. This dataset included four GBM samples, enabling the spatial localization of gene expression within the tumor architecture.

scRNA-seq dataset analysis

The data were imported into R (version 4.4.2) and analyzed with the Seurat R package (version 5.3.0) [18]. We performed quality control on the data to remove low-quality samples. The criteria were set as: (1) mitochondrial gene expression not exceeding 10% (2) minGene = 300, maxGene = 7500. Then we used the "DoubletFinder" function to eliminate the effect of doublets, and the impact of cell cycle was also regressed out. Then we used the "NormalizeData" function to normalize the data and the "FindVariableGenes" function to identify 2000 highly variable genes. "Harmony" was applied to perform dimensionality reduction, clustering, and batch effect correction [19]. Subsequently, the "Run PCA" function and "FindNeighbors" function were used to determine the optimal K value. Then, an appropriate resolution was selected for visualization, and the "Dimplot" function was used to plot the UMAP. The "inferCNV" R package was used to annotate and distinguish normal and tumor cells. The "scCATCH" R package was used to annotate normal cells, and the "Celltypist" function was used to annotate immune cells. The "CellChat" R package was used to analyze cell-cell interaction. The "CytoTRACE" R package was used to analyze cell stemness. The "GSVA" R package and the ssGSEA algorithm were used to evaluate the lactate score of neoplastic cells. The lactate-related genes (LRGs) based on the published studies, and a total of 332 genes were included in this study (Table S2) [20, 21]. The "SCENIC" R package was used to analyze transcription factor activity. Single-cell pseudotime trajectories were analyzed using the R package "monocle".

Spatial transcriptomics analysis

The Seurat R package was used to process and interpret spatial transcriptomics data. Data were normalized using the SCTransform function in Seurat. Then we use the "spacexr" R package to perform Robust Cell Type Decomposition (RCTD) deconvolution for spot annotation. Cell-cell communication analysis was conducted using the "Cellchat" package. The "squidpy" package was used to perform neighborhood enrichment analysis. Transcription factor activity was inferred using the Univariate Linear Model (ULM) algorithm implemented in the "decoupleR" package. The SciPy spatial distance module and NumPy and Pandas in Python were used to perform spatial correlation coefficients and distance-based analysis. Visualization of group-wise distance differences was performed using Matplotlib.

Machine learning algorithms

First, univariate Cox (unicox) regression analysis was used to categorize the tumor cells based on their lactate scores. Then, perform differential expression gene (DEG) analysis to identify the corresponding set of differentially expressed genes. The R packages "ggplot2" and "ggVolcano" were used to plot the DEGs. Principal Component Analysis (PCA) was used for dimensionality reduction and to validate the grouping. Gene expression matrices and corresponding survival data were collected from TCGA, CGGA, GSE108474, GSE4412, and the meta-cohort. Samples with missing survival information were excluded. Genes with average expression values greater than zero and shared across all datasets were retained for downstream analyses. The R package "Mime1" was used to construct prognostic models using 101 machine learning algorithms [22, 23]. Finally, the models "StepCox[forward] + RSF" were selected, and the AUC values were calculated, and ROC analysis was performed across different datasets.

Transcriptome analysis

The TCGA dataset was used for transcriptomic analysis; we used machine-learning-derived TCGA grouping results. The R package "IOBR" was used to perform immune infiltration analysis [24]. The R packages "ComplexHeatmap" and "ggplot2" were used to draw a heatmap. The R packages "maftools" and "GISTIC" were used to perform copy number variation (CNV) analysis. Subsequently, Gene Set Enrichment Analysis (GSEA) was performed to identify pathways that were significantly enriched between the two groups. Then, the R package "PharmacoGx" was used for CMAP analysis [25]. After that, the R packages "ISOpureR", "impute", "pRRophetic", and "SimDesign" were used to perform drug target analysis and GDSC analysis [26, 27]. The R package "scMetabolism" was used to calculate the metabolic activity in single-cell samples.

Cell culture and transfection

The mouse GBM cell line, GL261 cells, was cultured in high-glucose Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS). GL261 cells were selected as a well-established murine GBM model widely used to study tumor progression and tumor-immune interactions. The mouse macrophage cell line, RAW 264.7 cells, was cultured in high-glucose DMEM supplemented with 10% FBS. RAW 264.7 cells were used as a representative macrophage model to investigate tumor-macrophage crosstalk. *NFATc4* and *SPP1* knockdown was achieved using small

interfering RNA (siRNA), and the transfection reagent was from Sangon Biotech (E607402). The *NFATc4*-siRNA sequences are 5'-UUGCUCGCUAGAAGUCCUC-3', 5'-UAUCCUCUGAGUCCAGCUC-3', 5'-UUAAGUUCAG ACUCCACCU-3'. The *SPP1*-siRNA sequences are 5'-A UCGUCAUCAUCGUCGUC-3', 5'-UUGGAGUGAAA GUGUCUGC-3', 5'-UAAAGCUUCUCCUCUGAGC-3'. The NC-siRNA sequence is 5'-UUCUCCGAACGUGUC ACGU-3'. Cells were seeded and cultured until reaching approximately 60% confluence at 24 h after plating. The transfection complexes were prepared with serum-free DMEM to achieve a final siRNA concentration of 20 nM. After 8 h of transfection, the medium was replaced with complete DMEM, and cells were continuously cultured for an additional 24–48 h before subsequent functional experiments. The knockdown efficiency of *NFATc4* and *SPP1* was validated by quantitative real-time PCR (RT-qPCR) and western blot. The mRNA and protein levels of NFATc4 and SPP1 were significantly reduced following siRNA transfection, with siRNA-1 and siRNA-2 exhibiting the highest knockdown efficiencies (over 60–70%). These validated siRNAs were subsequently used for all functional assays.

After being transfected with *NFATc4*-siRNA, GL261 cells were collected for the Transwell assay, EdU proliferation assay, apoptosis detection, and cell cycle detection to evaluate the functional impact of *NFATc4* silencing on tumor cell behavior. We also designed a co-cultured system to investigate the functional interaction between tumor cells and macrophages [28]. RAW 264.7 cells were transfected with *SPP1*-siRNA and then seeded in the upper chamber of a Transwell insert (Corning, 3428), while GL261 cells were cultured in the lower chamber. After co-culture, the GL261 cells in the lower chamber were collected specifically for subsequent EdU proliferation assay, apoptosis detection, and cell cycle detection. Conversely, RAW 264.7 cells were transfected with *SPP1*-siRNA and then seeded in the lower chamber of a Transwell insert (Corning, 3428), while GL261 cells were cultured in the upper chamber. After co-culture, the GL261 cells in the upper chamber were subjected to the Transwell assay. In addition, GL261 cells were transfected with *NFATc4*-siRNA and then seeded in the upper chamber of a Transwell insert (Corning, 3428), while RAW 264.7 cells were cultured in the lower chamber. After co-culture, the RAW 264.7 cells in the lower chamber were collected specifically for subsequent macrophage polarization detection and western blot assay. Conversely, GL261 cells were transfected with *NFATc4*-siRNA and then seeded in the lower chamber of a Transwell insert (Corning, 3428), while RAW 264.7 cells were cultured in the upper chamber. After co-culture, the RAW 264.7 cells in the upper chamber were subjected to the Transwell assay. The mRNA and protein levels of NFATc4 were explored after adding 50 mM sodium lactate (HY-B2227B, MedChemExpress, USA) to GL261 cells for 24 h.

Western blot assay

To extract total protein, cells were lysed and collected in RIPA buffer containing protease and phosphatase inhibitors. SPP1 antibody (30200-1-AP, Rabbit, Proteintech, China, 1:1000) and NFATc4 antibody (ab99431, Rabbit, Abcam, USA, 1:2000) were used as the primary antibodies. β-actin (66009-1-Ig, Mouse, Proteintech, China, 1:5000) was used as the loading control. HRP-conjugated goat anti-Mouse IgG (H + L) (SA00001-1, Proteintech, China, 1:5000) and HRP-conjugated goat anti-Rabbit IgG (H + L) (SA00001-2, Proteintech, China, 1:5000) were used as the secondary antibody. Proteins were detected and quantified using a High-Sensitivity ECL Chemiluminescence Detection Kit (P2100, NCM Biotech, China) on a Bio-Rad ChemiDoc Imaging System.

RT-qPCR

Total RNA was isolated using the TRIzol Reagent (Invitrogen, 15596018CN). For cDNA synthesis, 1 μg of total RNA per sample was reverse transcribed using the Prime Script Kit (Accurate Biology, AG11705) following the manufacturer's protocol. Real-time PCR was performed in triplicate using a SYBR Green fluorescent-based assay (Accurate Biology, AG11701). Relative mRNA expression levels were calculated using the 2^−(ΔΔCt) method, normalized to GAPDH as the internal control. The calculations were as follows: ΔCt = Ct (target gene) − Ct (GAPDH) and ΔΔCt = ΔCt (case) − ΔCt (control). The primer sequences are provided in Table S1.

EdU proliferation assay

Cell proliferation was assessed using an EdU assay kit (Beyotime, C0071S). GL261 cells were incubated with the EdU reagent, fixed with 4% paraformaldehyde, and then subjected to fluorescent labeling of the incorporated EdU. Nuclei were counterstained with DAPI, and Edu-positive cells were visualized by fluorescence microscopy.

Flow cytometry for apoptosis detection

Cell apoptosis was analyzed using an Annexin V-APC apoptosis detection kit (KeyGEN BioTECH, KGA1030). GL261 cells were harvested, washed with cold phosphate-buffered saline, and resuspended in binding buffer. GL261 cells were then stained with Annexin V-APC and propidium iodide in the dark, and the number of apoptotic cells was quantified by flow cytometry.

Flow cytometry for cell cycle detection

GL261 cells were collected, washed with phosphate-buffered saline, and fixed in cold 75% ethanol at 4℃

overnight. After fixation, GL261 cells were incubated with propidium iodide in the dark, and the proportions of cells in the G0/G1, S, and G2/M phases were quantified by flow cytometry.

Transwell assay

GL261 cells were first resuspended in serum-free medium. 200 μL of cell suspension was added to the upper chamber of a Transwell insert (Corning, 3428). The lower chamber was supplemented with 700 μL of complete medium containing 10% FBS. The plate was incubated for 48 h at 37 °C under 5% CO_2. The cells that had traversed the membrane were fixed with 4% paraformaldehyde and stained with crystal violet.

In the co-culture system, RAW 264.7 cells were resuspended at a density of 1×10^5/ml and were added to the lower chamber. GL261 cells were resuspended at a density of 1×10^5/ml, and a 100 μL suspension of GL261 cells was added to the upper chamber. After coculturing for 48 h, GL261 cells that had traversed the membrane were fixed with 4% paraformaldehyde and stained with crystal violet. Likewise, GL261 cells were resuspended at a density of 1×10^5/mL and were added to the lower chamber. RAW 264.7 cells were resuspended at a density of 1×10^5/ml, and a 100 μL suspension of RAW 264.7 cells was added to the upper chamber. After coculturing for 48 h, RAW 264.7 cells that had traversed the membrane were fixed with 4% paraformaldehyde and stained with crystal violet.

Flow cytometry for macrophage polarization detection

RAW 264.7 cells were resuspended in 500 μL intracellular fixation buffer for 30 min. After centrifugation, RAW 264.7 cells were resuspended in 1 mL permeabilization buffer. After centrifugation again, RAW 264.7 cells were resuspended in 100 μL permeabilization buffer and incubated with anti-F4/80 (11-4801-82, eBioscience, USA) and anti-CD206 (12-2061-82, eBioscience, USA) antibodies in the dark for 30 min. Cells were then washed three times with staining buffer, resuspended in 200 μL staining buffer, and subjected to flow cytometric analysis.

Statistical analysis

All in vitro functional experiments were performed with 3 independent biological replicates. Statistical analyses were performed using GraphPad Prism. The data is presented as mean ± SD. For normally distributed variables, Student's two-tailed t-test was used for comparisons between two groups, and the one-way ANOVA test was used for comparisons between more than two groups. Survival curves were analyzed using Kaplan-Meier analysis and the log-rank test. In addition, the Wilcoxon rank-sum test was used to compare the area under the curve (AUC) values between high- and low-risk score groups in drug sensitivity analyses (CTRP and PRISM datasets). Correlation analyses between drug response and risk score were performed using Pearson correlation coefficients.

Results

Single-cell transcriptomic profiling identifies major cellular components of the GBM microenvironment

We analyzed scRNA-seq data from GSE138794, comprising 12,618 high-quality cells after rigorous quality control and doublet removal. Unsupervised clustering resolved 24 distinct clusters, which we annotated into 10 major cell types: neoplastic cells, macrophages, T cells, astrocytes, oligodendrocytes, stem cells, neurons, endothelial cells, innate lymphoid cells (ILCs), and fibroblasts (Fig. 2A). The Uniform Manifold Approximation and Projection (UMAP) plot showed the distribution of the cell types of the scRNA-seq data (Fig. 2A). Cell cycle analysis revealed that the majority of neoplastic cells resided in the G1 phase, aligning with their heightened metabolic activity. InferCNV analysis revealed widespread chromosomal copy number alterations in neoplastic cells compared with normal reference cells, supporting the malignant characteristics and genomic instability of the tumor population (Fig. 2B). Cell-cell communication analysis using CellChat highlighted frequent and strong interactions between neoplastic cells and macrophages (Fig. 2C–D). The proportion of 10 major cell types is shown in (Fig. 2E). CytoTRACE analysis indicated that oligodendrocytes and fibroblasts possessed the highest differentiation potential, whereas neoplastic cells and macrophages displayed lower scores, suggesting a more differentiated state (Fig. 2F).

PySCENIC analysis identifies NFATc4 as a key transcription factor associated with high-lactate neoplastic cells

Using the ssGSEA algorithm and a curated lactate-related gene set, we calculated lactate metabolic scores for neoplastic cells and stratified them into high- and low-lactate subgroups. Notably, the lactate score showed inconsistent distribution patterns with glycolysis- and hypoxia-related signatures, supporting the biological relevance and specificity of the lactate-related signature in GBM (Fig. S1) [29, 30]. PySCENIC analysis revealed distinct transcription factor (TF) regulon activities between the two groups (Fig. 3A). TF modules were visualized in a cluster heatmap and UMAP, highlighting six coherent regulon groups (Fig. 3B). Differential TF activity analysis identified SPI1, NFATc4, HOXC9, POU3F3, and FOXO1 as significantly elevated in high-lactate neoplastic cells (Fig. 3C). Survival analysis across six independent cohorts (CGGA693, CGGA625, CGGA301, GSE108474, GSE4412, and TCGA) showed that high NFATc4 expression was generally associated with poor patient prognosis

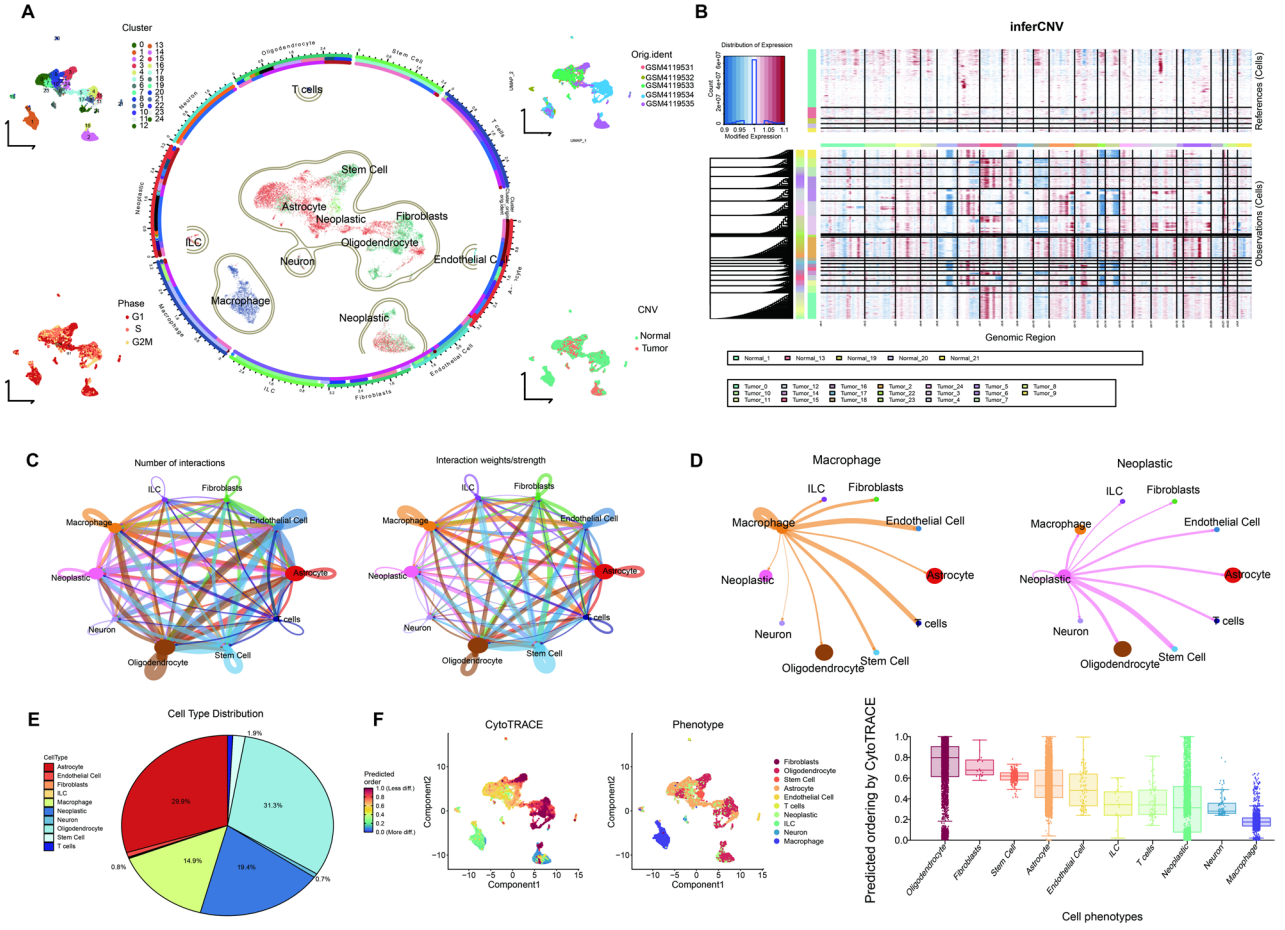

Fig. 2 Single-cell transcriptomic characterization of the GBM microenvironment. (**A**) UMAP visualization of 12,618 single cells colored by annotated cell types. (**B**) inferCNV analysis distinguishing neoplastic cells from normal cells. (**C**–**D**) CellChat analysis depicting the number and strength of intercellular interactions, with emphasis on neoplastic cell–macrophage communication. (**E**) Stacked bar plot showing the proportional composition of major cell types. (**F**) CytoTRACE scores projected onto UMAP, indicating the potential for stemness across cell populations

across most cohorts (Fig. 3D). NFATc4 regulatory networks and module distributions were further delineated (Fig. 3E–F). UMAP visualization confirmed elevated NFATc4 regulon activity specifically in high-lactate regions (Fig. 3G). A bubble plot summarized the differences in TF activity between metabolic subgroups (Fig. 3H).

Pseudotime trajectory analysis reveals dynamic NFATc4 upregulation during neoplastic cell differentiation

Pseudotime analysis of neoplastic cells using Monocle identified two branch points and five distinct cellular states (Fig. S2A–C). NFATc4 expression increased markedly along one branch after the second branch point, suggesting a role in promoting invasive phenotypes (Fig. S2D). Branch-dependent expression analysis revealed that AHCYL1 expression peaked and then declined. In contrast, CADPS showed sustained upregulation along the same branch, a pattern associated with unfavorable clinical outcomes at later differentiation stages (Fig. S2E). GO enrichment analysis of state-dependent genes highlighted terms such as "cell death" and "gliogenesis" (Fig. S2F). A heatmap displayed the expression patterns of the top 100 branch-dependent genes for the second branch (Fig. S2G).

Spatial transcriptomics reveals colocalization and interaction between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages

Spatial transcriptomics data (GSE194329, encompassing 11,970 spots) were deconvoluted to map lactate metabolic activity. NFATc4 expression across tissue regions (classified as high-lactate, low-lactate, and non-tumor areas) in GBM spatial transcriptomic slides (GSM5833533 and GSM5833534) is shown in Fig. 4A. Spatial colocalization analysis confirmed the proximity between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells within the tumor microenvironment in GBM spatial transcriptomic slides (GSM5833533 and GSM5833534), suggesting potential paracrine crosstalk

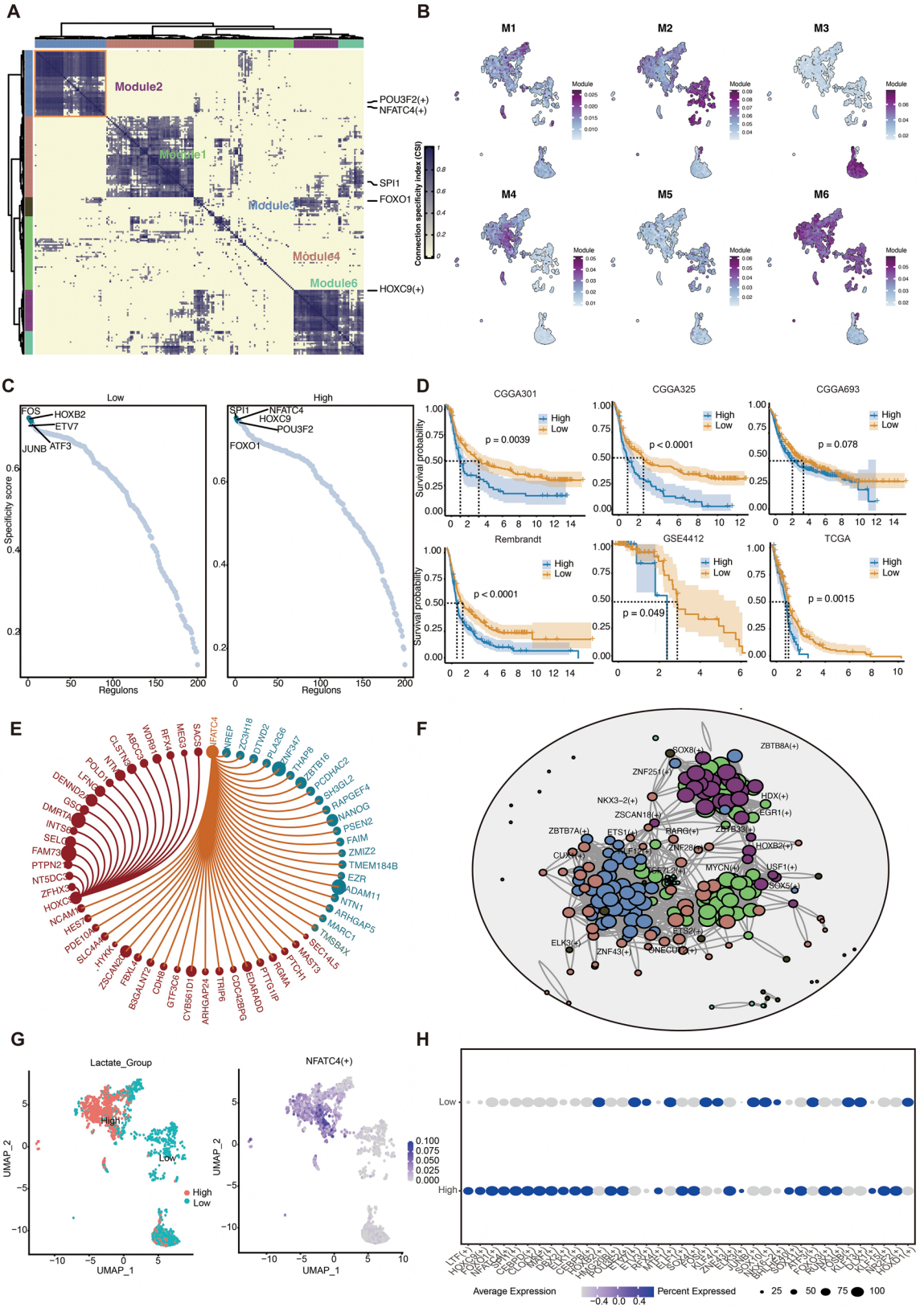

Fig. 3 (See legend on next page.)

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Fig. 3 Transcription factor activity analysis reveals NFATc4 as a key transcription factor associated with high lactate metabolism. (**A**) Heatmap of transcription factor modules identified by pySCENIC. (**B**) UMAP projection of regulon activity across six TF modules. (**C**) Differential TF expression between high- and low-lactate neoplastic cells. (**D**) Kaplan–Meier survival analysis of NFATc4 expression-based groups in GBM cohorts. (**E**) Regulatory network of NFATc4 and its target genes. (**F**) Regulon-based clustering of TF activities. (**G**) UMAP showing NFATc4 activity stratified by lactate metabolic group in neoplastic cells. (**H**) Bubble plot highlighting differentially active TFs between metabolic subgroups

(Fig. 4B). Besides, macrophage-enriched regions in GBM spatial transcriptomic slides (GSM5833533 and GSM5833534) are shown in Fig. 4C. CellChat analysis of spatially resolved data identified strong interaction signals between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells (Fig. 4D–E). The spatial transcriptomic analysis in the remaining GBM spatial transcriptomic slides (GSM5833535 and GSM5833536) is presented in the Supplementary Materials (Fig. S3). To further quantify the spatial relationship between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages, we performed distance-based spatial analyses and neighborhood enrichment analysis. The results showed significant enrichment observed between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells, supporting their spatial proximity within the tumor microenvironment (Fig. 4F). In addition, distance distribution analysis demonstrated an accumulation of short-range interactions between these two populations, indicating that NFATc4⁺ neoplastic cells are preferentially localized in proximity to SPP1⁺macrophages rather than being randomly distributed (Fig. 4G). The overall nearest spatial proportion of NFATc4⁺ neoplastic cells was located in proximity to SPP1⁺ macrophages (Fig. 4H). To further assess the impact of lactate metabolic status on their spatial organization, mean nearest-neighbor distances were compared between the high and low lactate groups, revealing reduced spatial separation under high-lactate conditions (Fig. 4I). Consistently, density distribution analysis demonstrated an enrichment of short-range interactions between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells in high-lactate regions compared with low-lactate regions (Fig. 4J). To further validate our findings, we calculated lactate metabolic scores using transcriptomic data from the GBM patient-derived xenograft (PDX) model of the Mayo Clinic, which was retrieved from the cBioPortal database (cBioPortal, https://www.cbioportal.org) (Fig. S4) [31]. Survival analysis of the GBM PDX model revealed that high NFATc4 expression was significantly associated with worse prognosis (Fig. S4A). Using signature genes of SPP1⁺ macrophages and NFATc4⁺ malignant cells identified via the FindMarkers tool, we further quantified their relative abundance in GBM PDX models by ssGSEA analysis. Intriguingly, a strong positive correlation was observed between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells (Fig. S4B). Additionally, GBM PDX models with higher lactate metabolic scores exhibited significantly elevated expression levels of both NFATc4 and SPP1 (Fig. S4C). Together, these findings further support enhanced spatial colocalization between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages within lactate-enriched GBM niches.

In vitro validation further revealed the mechanisms between the NFATc4⁺ neoplastic cells and the SPP1⁺ macrophages in the TME

Prior bioinformatic screening identified NFATc4 and SPP1 as signature genes specifically enriched in neoplastic cells and macrophages, respectively. Integrative single-cell RNA-seq and spatial transcriptomic analyses further revealed intimate spatial and molecular linkage between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages. To dissect their functional interplay, we performed a series of in vitro loss-of-function and co-culture experiments.

First, we validated the knockdown efficiency of three independent siRNAs targeting *Nfatc4* and *Spp1*. In GL261 GBM cells, all three *Nfatc4*-targeting siRNAs markedly reduced *Nfatc4* transcript levels (Fig. 5A); among these, siRNA-*Nfatc4*-1 and siRNA-*Nfatc4*-2 exhibited robust suppression of NFATc4 protein expression and were selected for subsequent functional assays (Fig. 5C). Similarly, three *Spp1*-specific siRNAs were screened in macrophages, with prominent silencing of *Spp1* transcription confirmed for all candidates (Fig. 5B). siRNA-*Spp1*-1 and siRNA-*Spp1*-2 were further verified to efficiently downregulate SPP1 protein abundance in macrophages and used for follow-up experiments (Fig. 5D).

EdU proliferation assays demonstrated that genetic silencing of *Nfatc4* significantly attenuated the proliferative capacity of GL261 cells relative to control siRNA groups (Fig. 5E and I). Flow cytometric apoptosis analysis showed that *Nfatc4* knockdown markedly increased the apoptotic population of GL261 cells (Fig. 5F and J). Cell cycle profiling revealed that *Nfatc4* silencing induced cell cycle arrest at the G1 and G2 phases, accompanied by a remarkable reduction in the S-phase cell fraction (Fig. 5G and K). Transwell migration assays further validated that NFATc4 silencing profoundly impaired the migratory potential of GL261 cells (Fig. 5H and L). Collectively, these data indicate that NFATc4 drives GBM cell proliferation, inhibits apoptosis, and promotes cell migration in vitro.

We next employed a co-culture system to explore the reciprocal crosstalk between neoplastic cells and SPP1⁺ macrophages. Macrophage migration was significantly compromised when co-cultured with *Nfatc4*-silenced

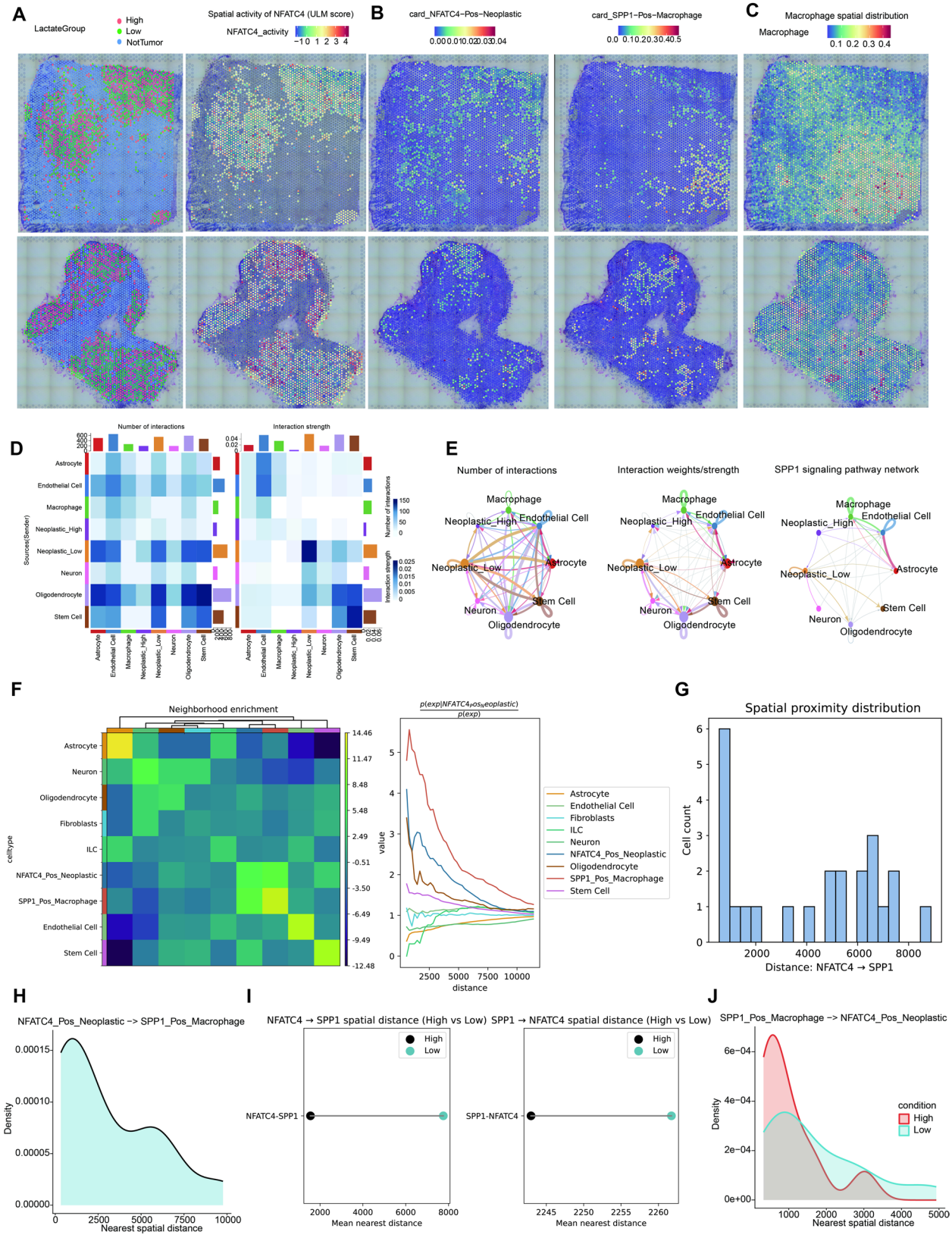

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Fig. 4 Spatial transcriptomics reveals colocalization of NFATc4⁺ neoplastic cells with SPP1⁺ macrophages. (**A**) Spatial distribution of lactate metabolic groups and NFATc4 activity scores in glioma tissues (**B**) Spatial proximity analysis indicates significant colocalization between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages. (**C**) Macrophage infiltration patterns demonstrate their regional colocalization. (**D**–**E**) CellChat interaction plots quantifying communication strength among spatially resolved cell types. (**F**) Neighborhood enrichment and spatial distance analyses indicate preferential spatial association between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages. (**G**-**J**) Spatial proximity and nearest-neighbor distance analyses further confirm close localization between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages, particularly in the high-lactate group

GL261 cells (Fig. 5M and O). Flow cytometric quantification showed that the proportion of pro-tumorigenic F4/80⁺CD206⁺ M2-like macrophages was markedly decreased upon *Nfatc4* knockdown in GBM cells (Fig. 5N and P). Consistently, SPP1 protein expression in macrophage was substantially downregulated following co-culture with *Nfatc4*-deficient GL261 cells (Fig. 5Q). We further tested the regulatory effect of lactate on NFATc4 expression. Exogenous lactate supplementation significantly elevated both *Nfatc4* mRNA and protein levels in GL261 cells (Fig. 5R and S), implying a positive feedback loop linking lactate metabolism to NFATc4 activation in GBM cells.

To further define the functional role of SPP1⁺ macrophages in modulating neoplastic cell behaviors, we knocked down *Spp1* in macrophages and performed co-culture with GL261 cells. EdU assays showed that *Spp1* silencing in macrophages significantly suppressed the proliferation of co-cultured GBM cells (Fig. 6A and E). Flow cytometry revealed elevated apoptosis in GL261 cells upon macrophage *Spp1* silencing (Fig. 6B and F). Cell cycle analysis demonstrated that loss of macrophage SPP1 triggered G1/G2-phase arrest and reduced the S-phase population in neoplastic cells (Fig. 6C and G). Moreover, Transwell assays confirmed that silencing *Spp1* in macrophages markedly reduced the migratory capacity of co-cultured GL261 cells (Fig. 6D and H).

In summary, these in vitro functional assays found a critical NFATc4–SPP1 signaling axis that mediates bidirectional crosstalk between neoplastic cells and macrophages, synergistically sustaining malignant progression within the glioma tumor microenvironment.

A machine learning-derived prognostic model based on lactate-associated genes

Building upon our integrative analyses of single-cell and spatial transcriptomics, together with in vitro experimental validation, we revealed a lactate-driven tumor-macrophage interaction network. We next sought to evaluate the prognostic relevance of lactate metabolic heterogeneity in larger clinical cohorts. To this end, we constructed a prognostic model based on lactate-associated genes using a comprehensive machine learning framework. Consensus clustering based on the LRGs identified two molecular subtypes in the TCGA cohort. Differential expression analysis between the resulting clusters (Cluster A and Cluster B) yielded a prognostic gene signature ($|\log_2 FC| > 1$, $P < 0.01$). A volcano plot was generated to visualize the differential expression patterns between the two clusters (Fig. 7B). PCA of TCGA samples stratified by this signature showed clear separation between clusters, confirming its discriminative power (Fig. 7C). Survival analysis between the two clusters further supported the prognostic relevance of the identified gene signature (Fig. 7D). To improve robustness across datasets, only shared non-low-expression genes across TCGA, CGGA, GSE108474, GSE4412 and the meta-cohort were retained for downstream modeling. Candidate genes were further filtered using univariate Cox regression in the TCGA training cohort with a stringent cutoff ($P < 0.001$). Subsequently, 101 machine-learning algorithm combinations were systematically evaluated, and the "StepCox[forward] + RSF" model was selected based on performance (Fig. 7A). The model achieved a C-index of 0.87 in the TCGA training set, and achieved the C-index of 0.64 and 0.62 in GSE4412 and the meta-cohort. Time-dependent ROC analysis demonstrated strong predictive accuracy, with AUCs reaching 0.976 and 0.905 at 5 years in TCGA and GSE4412, respectively (Fig. 7E). Meta-analysis of univariate Cox regression results across datasets yielded a pooled hazard ratio (HR) of 1.82 (95% CI: 1.64 – 2.02, *p* < 0.001), confirming the signature's association with poorer survival (Fig. 7F). The calibration curves and the 1-year decision curve analysis also provided in Fig. S5. The calibration curves indicated good consistency between predicted and observed 1-year survival probabilities across these cohorts. The DCA curves demonstrated the favorable clinical utility of the risk model (Fig. S5).

High-risk group exhibits distinct immune infiltration and genomic alteration profiles

Immune deconvolution using multiple algorithms (MCP-counter, EPIC, xCell, CIBERSORT, etc.) revealed significantly elevated infiltration of macrophages, CD8⁺ T cells, and NK cells in the high-risk group (Fig. 8A). Mutational profiling identified missense mutations as the most common variant class, with TTN, TP53, PTEN, and EGFR as frequently altered genes (Fig. 8B). Maftools analysis of the high-risk group further revealed that 97.7% of samples harbored somatic mutations, with PTEN, EGFR, TP53, and TTN being the most commonly altered genes (Fig. 8C). CNV analysis showed more frequent and extensive genomic gains/losses in high-risk samples,

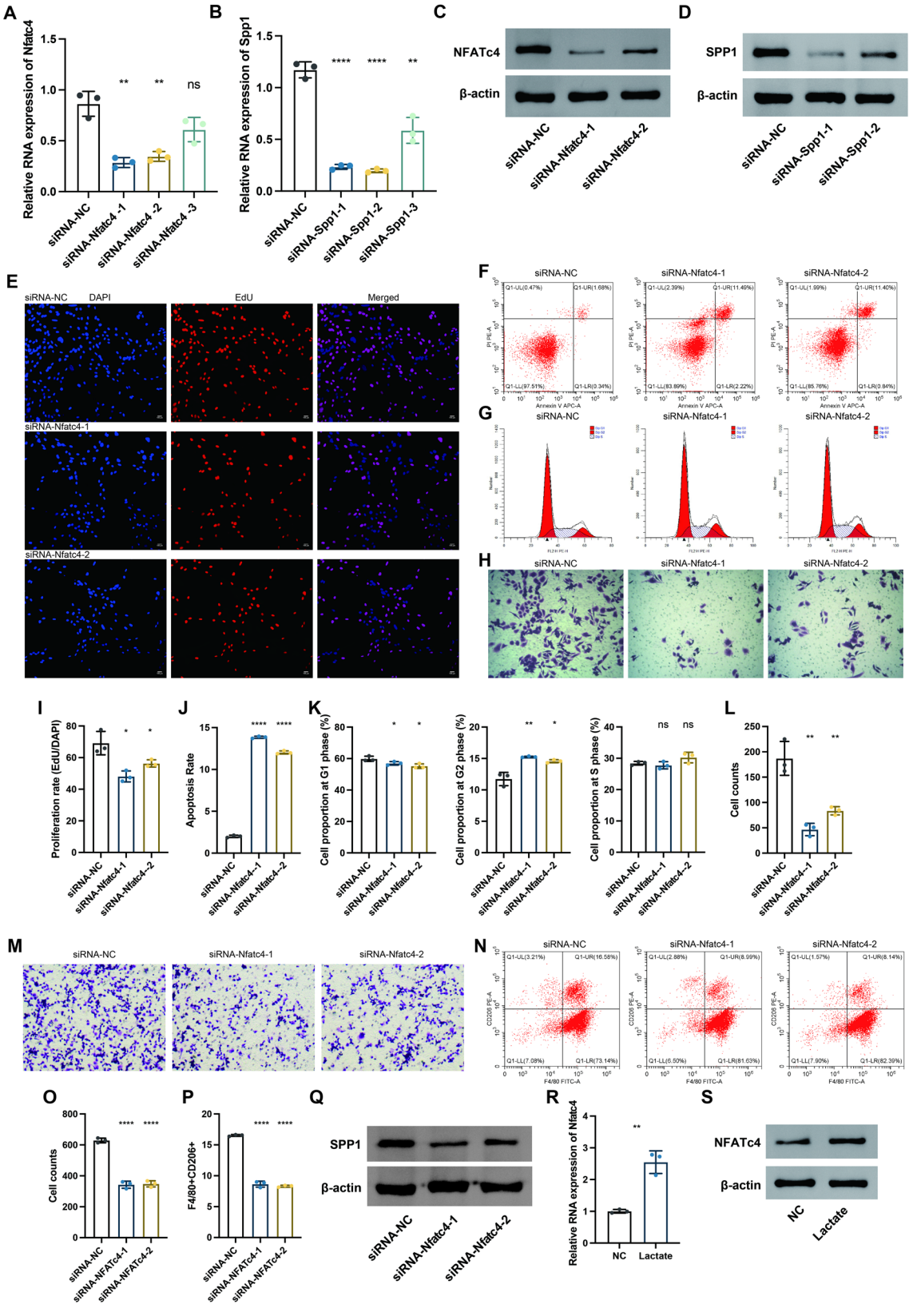

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Fig. 5 In vitro experiments demonstrated the functional involvement of NFATc4 in the crosstalk between GBM cells and macrophages. (**A**-**B**) qRT-PCR validation of knockdown efficiency for three independent siRNAs targeting *Nfatc4* in GL261 cells and *Spp1* in macrophages. (**C**-**D**) Western blot analysis confirming efficient suppression of NFATc4 and SPP1 protein expression following siRNA transfection. (**E** and **I**) EdU incorporation assays demonstrating reduced proliferative capacity of GL261 cells after *Nftac4* silencing. (**F** and **J**) Flow cytometric apoptosis analysis showing increased apoptosis in Nfatc4-deficient GL261 cells. (**G** and **K**) Cell cycle analysis revealing G1/G2 phase arrest and reduced S-phase proportion following *Nfatc4* knockdown. (**H** and **L**) Transwell migration assays indicate impaired migratory ability of GL261 cells after Nftac4 silencing. (**M** and **O**) Co-culture migration assays showing reduced macrophage recruitment induced by *Nfatc4*-silenced GL261 cells. (**N** and **P**) Flow cytometry analysis demonstrating decreased proportions of $F4/80^+$ $CD206^+$ M2-like macrophages upon co-culture with Nfatc4-deficient GBM cells. (**Q**) Western blot analysis showing reduced SPP1 expression in macrophages after co-culture with *Nfatc4*-silenced GL261 cells. (**R** and **S**) qRT-PCR and western blot analyses indicate that lactate treatment upregulates NFATc4 expression in GL261 cells. All experiments were performed with 3 independent biological replicates. Data are presented as mean ± SD. *$p < 0.05$, **$p < 0.01$, ****$p < 0.0001$, ns: not significant

particularly on chromosomes 1, 7, and 9 (Fig. 8D). GSVA enrichment analysis indicated upregulation of immune-related pathways (e.g., leukocyte migration, cytokine signaling) and downregulation of differentiation and DNA recombination pathways in high-risk tumors (Fig. 8E). These findings were consistent with our earlier analyses, where a high lactate state was associated with increased macrophage infiltration, reflecting the interaction between $NFATc4^+$ neoplastic cells and $SPP1^+$ macrophages.

Drug sensitivity and metabolic pathway activities associated with risk stratification

CMAP_XSum analysis identified several small molecules (e.g., fasudil, imatinib) as potentially effective in high-risk patients (Fig. 9A). Drug-target correlation analysis revealed a negative association between the risk score and the estimated IC_{50} for multiple targeted agents (Fig. 9B–C). OncoPredict analysis further indicated significantly lower predicted IC_{50} values for A.443,654 and A.770,041 in the high-risk group (Fig. 9D). Single-cell metabolic scoring highlighted elevated glycolysis and oxidative phosphorylation activities in specific neoplastic subpopulations (Fig. 9E–F).

Discussion

GBM is a highly heterogeneous disease with complex interactions between tumor cells and the TME. Despite the remarkable advances in modern science, it remains challenging to fully elucidate the pathogenic mechanisms and develop effective therapeutic strategies [32–34]. The metabolism of tumor cells has emerged as a key area of research in recent years. This phenomenon is primarily due to the distinct metabolic reprogramming of tumor cells compared with normal cells, which may represent potential therapeutic targets. The most well-known effect of lactate metabolism in tumor cells is the Warburg effect, which enables tumor cells to obtain large amounts of energy. Nowadays, studies have demonstrated that lactate promotes tumor growth by inducing an acidic TME, thereby inhibiting CD8 + T-cell proliferation and cytokine secretion [35]. Meanwhile, lactate metabolism also promotes the function of immunosuppressive cells, such as upregulating PD-1 expression in regulatory T cells (Tregs), leading to diminished therapeutic responses, while simultaneously driving the differentiation of M2 macrophages [36, 37]. Given this pivotal role, we hypothesized that profiling neoplastic cells using lactate-related gene signatures could reveal their metabolic heterogeneity [21]. In this study, through scRNA-seq and spatial transcriptomics analyses, we identified the transcription factor NFATc4 as a key prognostic indicator specifically upregulated in neoplastic cells with vigorous lactate metabolism. Furthermore, our data revealed a potential symbiotic mechanism in which $NFATc4^+$ neoplastic cells actively interact with $SPP1^+$ macrophages to foster a poor-prognosis niche.

Our research first identified 10 cell types in the dataset GSE138794 and then used AUCell and LRGs to calculate lactate metabolic scores for neoplastic cells. Since upregulated metabolic genes are hallmarks of tumor aggressiveness, this scoring strategy allowed us to stratify neoplastic cells into high- and low-activity groups. To identify transcription factor differences between neoplastic cells with high and low lactate levels, we performed pySCENIC analysis and identified NFATc4 as the major transcription factor most closely associated with the high-lactate metabolic state. Accordingly, NFATc4 was markedly enriched in tumor cells with elevated lactate metabolism by in vitro validation. NFATc4 is a member of the NFAT family and has been proven to be associated with the progression of various cancers [15–17]. In our study, NFATc4 serves as a marker of poor prognosis in six GBM cohorts.

We next performed pseudotime analysis to find the differentiation trajectory of neoplastic cells. We observed that genes known to be associated with poor prognosis, such as AHCYL1 and CADPS, were highly expressed during the middle and late stages of differentiation [38–42]. Consistent with this finding, we found that during differentiation, NFATc4 activity gradually increased in later stages, suggesting that its expression is consistent with tumor progression. During the differentiation process, we found that NFATc4 was mainly enriched along the astrocyte differentiation branch, suggesting that neoplastic cells derived from astrocyte lineages exhibit

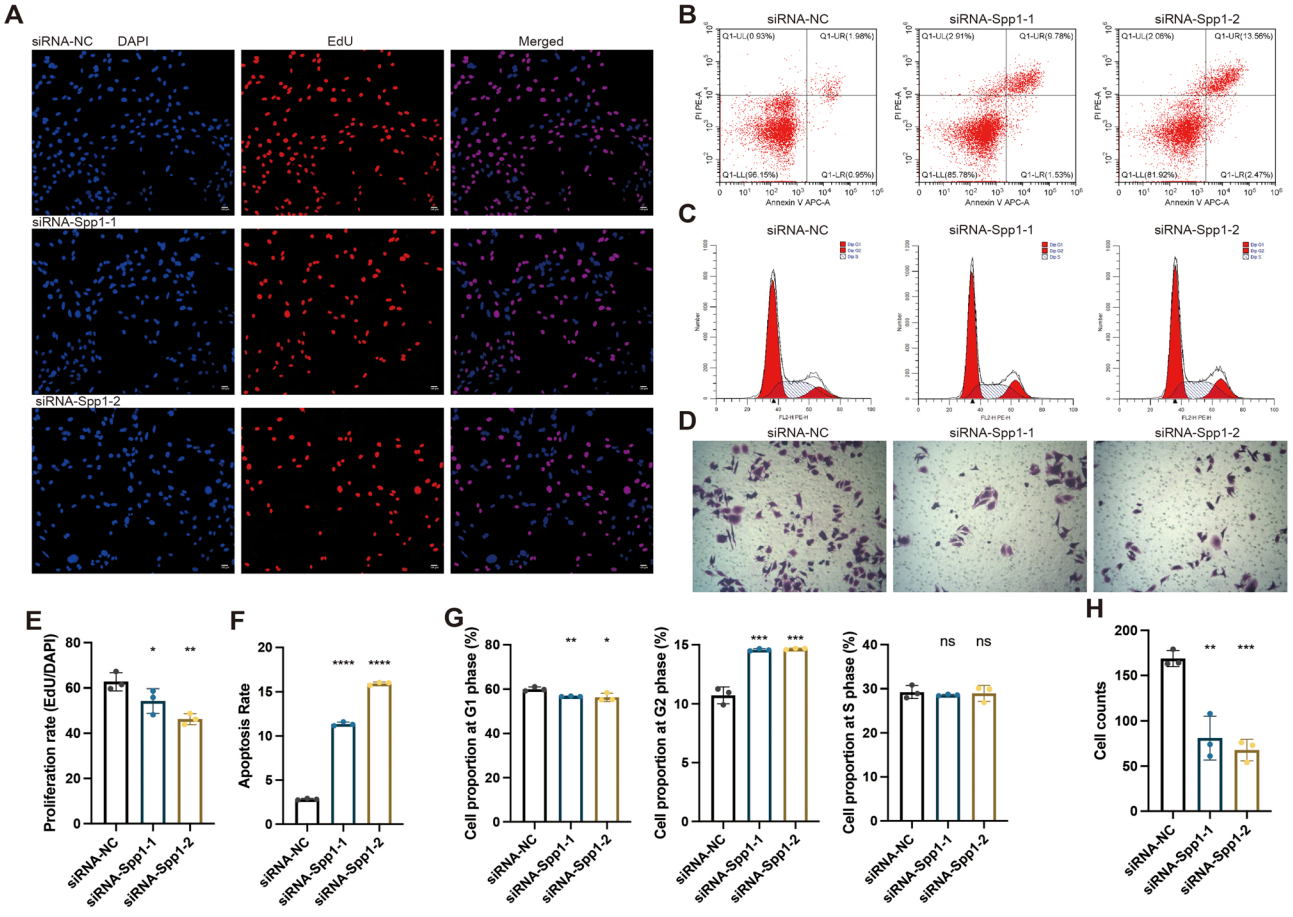

Fig. 6 In vitro experiments demonstrated the functional involvement of the SPP1 in the crosstalk between GBM cells and macrophages. (**A** and **E**) EdU incorporation assays showing that *Spp1* knockdown in macrophages significantly suppresses the proliferation of co-cultured GL261 GBM cells. (**B** and **F**) Flow cytometric apoptosis analysis demonstrating increased apoptosis in GBM cells following macrophage *Spp1* silencing. (**C** and **G**) Cell cycle analysis revealing G1/G2 phase arrest and decreased S-phase proportion in GL261 cells co-cultured with Spp1-deficient macrophages. (**D** and **H**) Transwell migration assays indicate that Spp1 silencing of macrophage markedly impairs the migratory ability of co-cultured GBM cells. All experiments were performed with 3 independent biological replicates. Data are presented as mean ± SD. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, ****$p < 0.0001$, ns: not significant

higher lactate metabolism. Astrocytes play a pivotal role in GBM growth, which can promote GBM progression by recruiting tumor-associated macrophages [43]. Therefore, we hypothesized that astrocytes, neoplastic cells, and macrophages interact within a high-lactate microenvironment to form a signaling network that drives GBM progression. To prove this, we next performed spatial transcriptomics analysis. The spatial transcriptomics study in the dataset GSE194329 revealed the spatial distribution of NFATc4 expression in tumor tissues, indicating that it is predominantly located in the tumor's central regions. In the subsequent CellChat analysis of NFATc4^{+} neoplastic cells, we found that they interacted prominently with SPP1^{+} macrophages, and the two cell types also showed a correlated spatial distribution. Importantly, our in vitro experiments provide compelling functional validation of the bioinformatically predicted interplay between NFATc4^{+} neoplastic cells and SPP1^{+} macrophages within the tumor microenvironment. The direct silencing of NFATc4 in GL261 cells robustly impairs core oncogenic phenotypes—proliferation, apoptosis, cell cycle, and migration—showing NFATc4 as a key intrinsic regulator of tumor aggressiveness. Crucially, the co-culture experiments demonstrate that this malignant capacity is extrinsically modulated by macrophages via SPP1, as silencing SPP1 specifically in macrophages recapitulates the same spectrum of inhibitory effects on neighboring tumor cells. This bidirectional functional linkage confirms that NFATc4-driven tumor cell fitness is not autonomous but is critically supported by a paracrine signaling axis from SPP1-expressing macrophages, highlighting a coordinated cellular crosstalk that promotes tumor progression. Past studies have elucidated the protumorigenic role of SPP1^{+} macrophages. For instance, in colorectal cancer (CRC), they promote tumor progression by interacting with FAP+ fibroblasts [44]. Moreover, it has been reported that SPP1^{+} macrophages promoted HNSCC progression by secreting TNF-α and IL-1β to

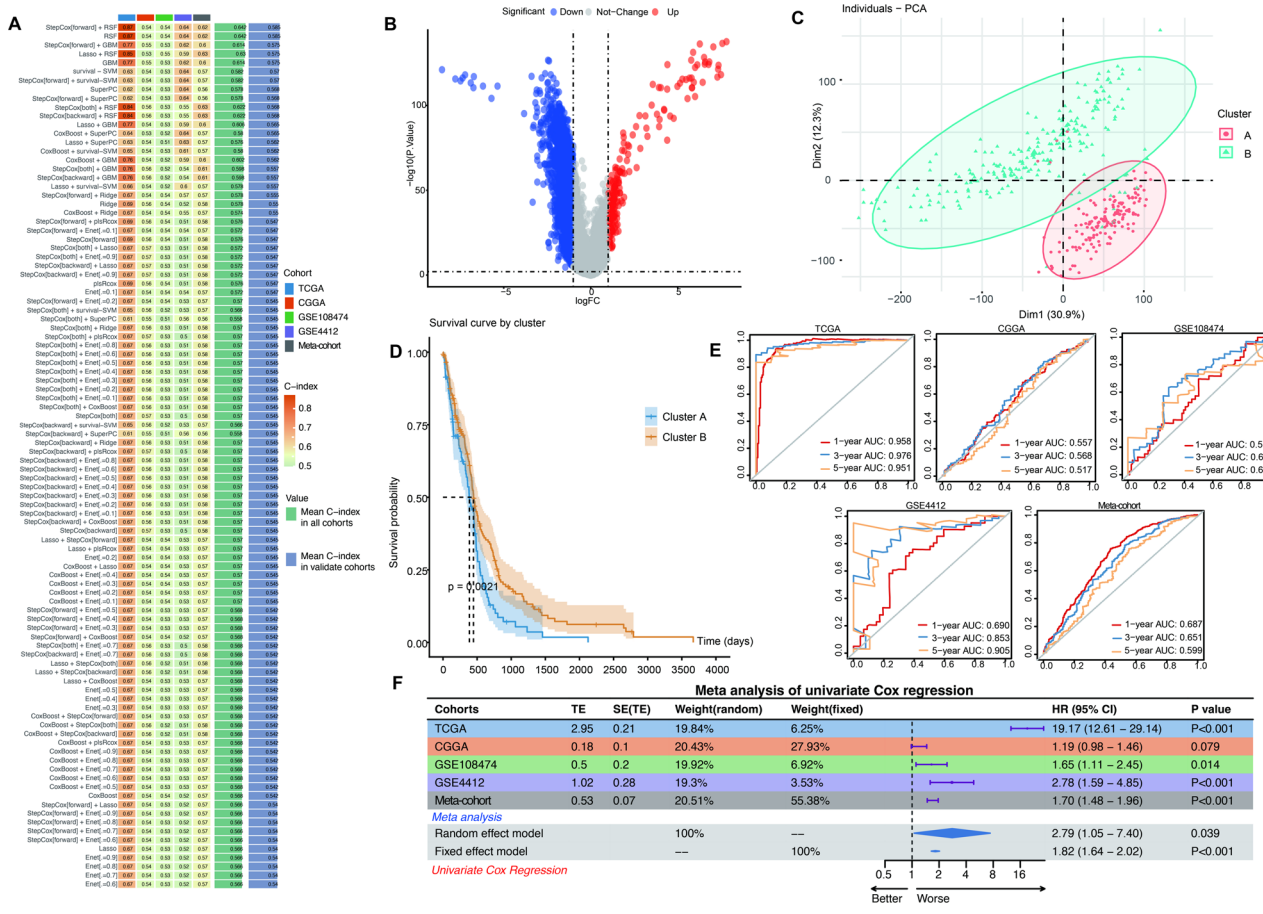

Fig. 7 Machine learning-based prognostic model for GBM. (**A**) Heatmap summarizing performance metrics of 101 machine-learning algorithm combinations. (**B**) Volcano plot of differentially expressed genes used for model construction. (**C**) PCA plot demonstrating distinct clustering of TCGA samples based on lactate-related gene signatures. (**D**) Survival analysis of the two clusters. (**E**) ROC curves showing AUC values at 1-, 3-, and 5-year survival in independent validation cohorts. (**F**) Forest plot of meta-analysis combining univariate Cox regression results across datasets

support tumor cell proliferation and migration [45]. These findings further support the correlation between SPP1⁺ macrophages and NFATc4⁺ neoplastic cells in GBM progression.

Importantly, this lactate-driven crosstalk between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages may underlie the adverse clinical features observed in our prognostic model. Specifically, this interaction promotes an immunosuppressive microenvironment, contributing to poorer survival in high-risk patients. We further explored the role of lactate metabolism in GBM using a machine-learning-based model in the datasets TCGA, CGGA, GSE108474, GSE4412, and meta-cohort. According to machine learning results, "StepCox[forward] + RSF" was identified as the best model and demonstrated strong predictive performance across other test sets, suggesting its potential application. Notably, we acknowledge that although the model showed strong performance in the training and certain validation cohorts, differences in cohort composition, intrinsic inter-tumor heterogeneity of GBM, and sequencing/platform variability may contribute to performance fluctuations across datasets. This also highlights the importance of external validation in the meta-cohort, which provides more reliable evidence for the model's generalizability. Moreover, while spatial transcriptomic and CellChat analyses suggest a close association between NFATc4⁺ neoplastic cells and SPP1⁺ macrophages, the current data are mainly correlative and do not fully establish a direct causal regulatory relationship. Our in vitro results provide partial functional support for this axis; however, more detailed mechanistic studies are required to further validate the NFATc4-SPP1 interaction. In addition, we acknowledge that our experimental validation was performed using mouse cell lines (GL261 and RAW 264.7), which may not fully recapitulate the complexity of human tumor-immune interactions. While these models are well-established for studying GBM biology and have provided valuable insights into tumor-macrophage crosstalk, we recognize that cross-species differences in gene expression, signaling pathways, and cellular phenotypes may affect the generalizability of our findings. Therefore,

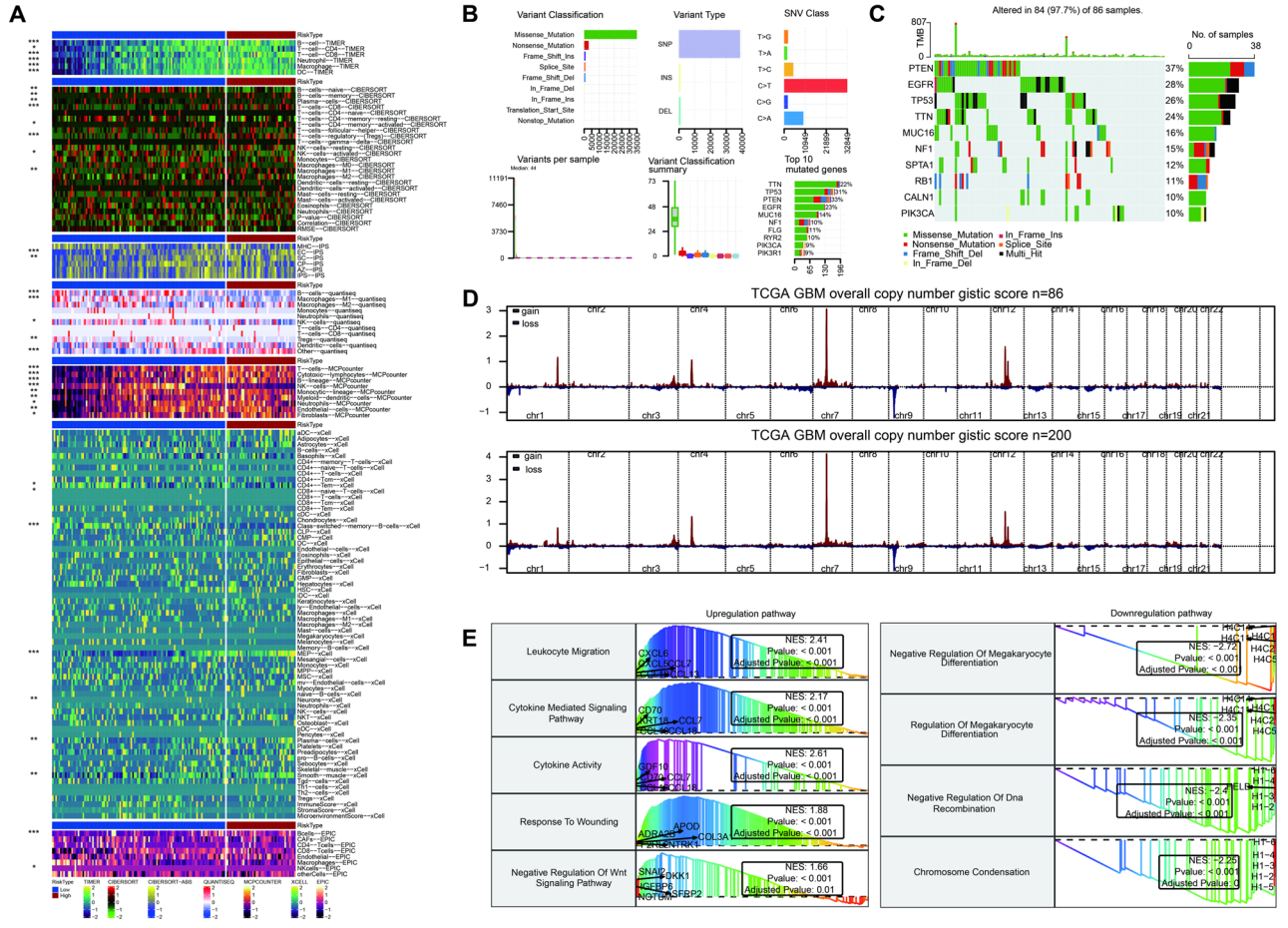

Fig. 8 Immunogenomic and mutational landscape of risk-stratified GBM. (**A**) Heatmap of immune cell infiltration estimated by multiple deconvolution algorithms in high- and low-risk groups. (**B**) Summary of mutational profiles: variant classification, SNV class, mutation burden, and top 10 mutated genes. (**C**) Oncoprint of recurrent mutations stratified by risk group. (**D**) Chromosomal copy number alteration profiles. (**E**) GSVA enrichment plot displaying upregulated and downregulated pathways in high-risk GBM. *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$

future validation in human cellular models is essential to confirm the translational relevance of the NFATc4-SPP1 axis identified in this study. In the subsequent immune infiltration analysis, the high-risk group showed higher scores for various immune cells, indicating increased infiltration and activity, which may be associated with the poor prognosis of GBM [46]. In the GSVA analysis, we observed upregulation and downregulation of relevant pathways in the high-risk group, including leukocyte migration, cytokine-mediated signaling, cytokine activity, response to wounding, and negative regulation of Wnt signaling. The enhanced cytokine signaling within the TME, along with tumor-associated tissue remodeling, may contribute to the immunosuppressive microenvironment. The negative regulation of Wnt signaling may reflect feedback mechanisms that counteract tumor proliferation or maintain tissue homeostasis under immune pressure. These pathway alterations indicate a dynamic interplay between neoplastic cells and the TME that may contribute to GBM progression; further studies are needed to clarify the mechanisms underlying this interplay.

As a pilot study, our research has certain limitations that should be noted and addressed in future work. First, the number of scRNA-seq samples included in our study is limited, which may affect the generalizability of our findings. Therefore, further validation in larger and more diverse cohorts is necessary to confirm the robustness of our results. Second, pySCENIC analysis identified several transcription factors (TFs) enriched in neoplastic cells with high lactate metabolism. While this study focuses on NFATc4 and its roles, the remaining TFs identified in the analysis warrant in-depth exploration in future research to fully elucidate their functions in GBM progression and in regulating lactate metabolism. Third, although we performed comprehensive bioinformatic analyses to predict the mechanism underlying the role of NFATc4$^+$ neoplastic cells in GBM, this proposed mechanism has not yet been experimentally validated. Notably, a key limitation of this study is the lack of in vivo validation using animal

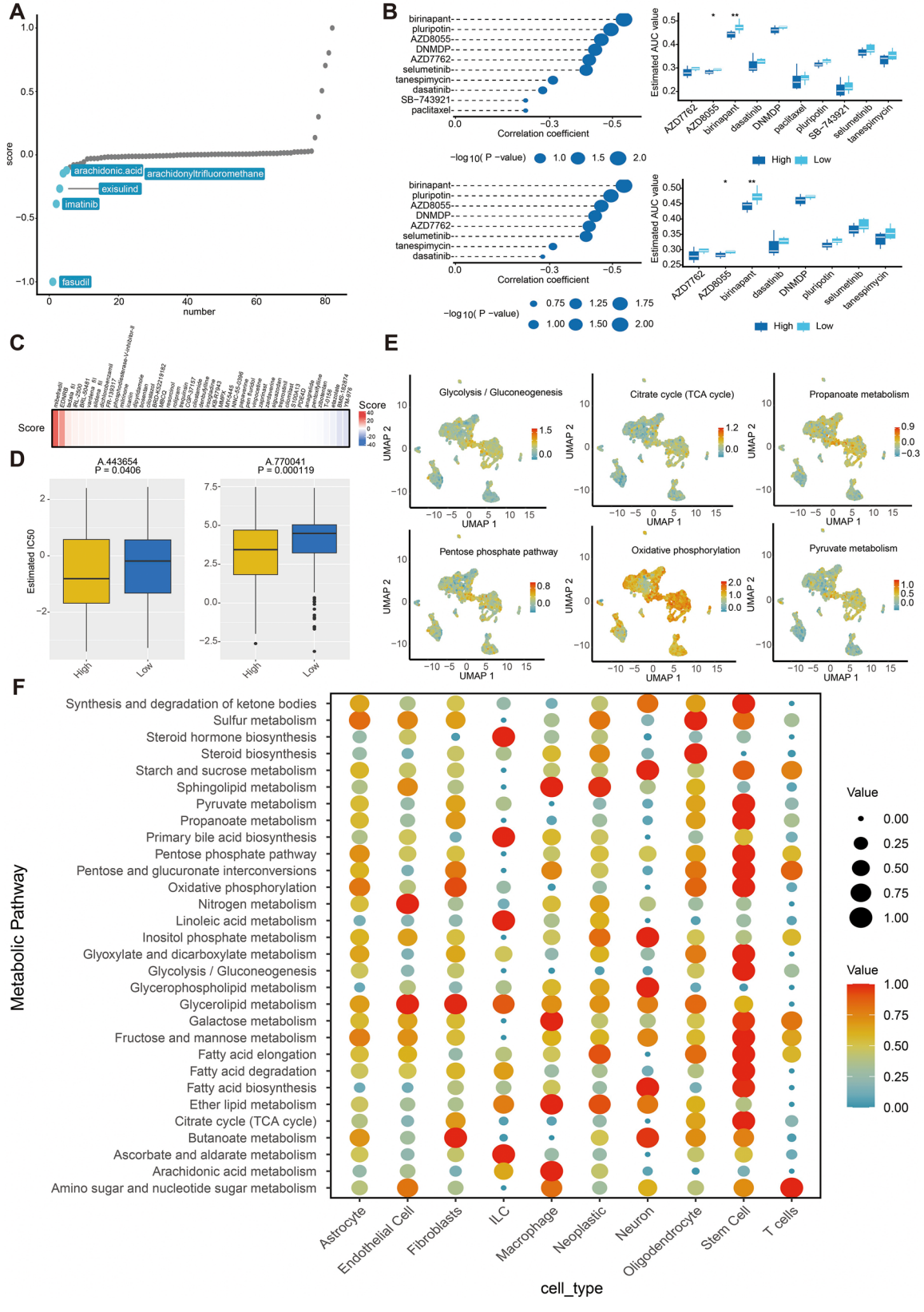

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Fig. 9 Therapeutic vulnerability and metabolic pathway activity in risk subgroups. (**A**) Top five candidate small-molecule compounds from CMAP_XSum analysis with predicted efficacy in high-risk patients. (**B**) Correlation between drug sensitivity (IC_{50}) and risk score for selected targeted agents. (**C**) Heatmap of drug response scores across risk groups. (**D**) Box plots comparing estimated IC_{50} values of representative compounds between risk groups. (**E**) UMAP visualization of metabolic pathway activity in single cells. (**F**) Scatter plot illustrating metabolic signaling intensity in high-risk associated pathways. * $p < 0.05$, ** $p < 0.01$

models. Animal experiments are essential to verify the in vivo function of NFATc4 and SPP1, as well as their interaction in the tumor microenvironment, which is critical for translating our in vitro findings into clinical relevance. Importantly, due to the retrospective nature of the publicly available datasets, clinical annotations such as IDH mutation status were not consistently available across cohorts. To maintain cohort consistency and avoid introducing additional stratification bias, we analyzed GBM (WHO grade IV) as a unified entity without further stratification based on IDH status. Finally, this study aims to explore the mechanisms linking lactate metabolism to GBM progression and further investigate its association with patient prognosis and treatment response. We hope that this pilot study will lay a foundation for future research and provide new insights into prognostic assessment and the development of related immunotherapeutic strategies for GBM patients.

Abbreviations

AUC	Area Under the Curve
C-index	Concordance Index
CNV	Copy Number Variation
CI	Confidence Interval
CRP	C-Reactive Protein
DEG	Differentially Expressed Gene
FDR	False Discovery Rate
GBM	Glioblastoma
GSEA	Gene Set Enrichment Analysis
HR	Hazard Ratio
KEGG	Kyoto Encyclopedia of Genes and Genomes
LRG	Lactate Related Gene
MGMT	O6-methylguanine-DNA methyltransferase
NK cells	Natural Killer cells
NFATc4	Nuclear Factor of Activated T-cells Cytoplasmic 4
PCA	Principal Component Analysis
PDX	Patient-derived xenograft
RNA-seq	RNA Sequencing
RSF	Random Survival Forest
ROC	Receiver Operating Characteristic
SPP1	Secreted Phosphoprotein 1
TAMs	Tumor-Associated Macrophages
TF	Transcription Factor
TME	Tumor Microenvironment
Tregs	Regulatory T cells
UMAP	Uniform Manifold Approximation and Projection
Wnt	Wingless/Integrated Pathway

Supplementary Information

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Supplementary Material 1

Supplementary Material 2

Supplementary Material 3

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Author contributions
HZ and ZX designed the study. RW and WW conducted the experiments. RW drafted the manuscript and drew the figures. XW, YY, ZZ, JC, YX, LZ, XY, HZ, and ZX revised the manuscript. HZ and ZX supervised the manuscript and provided funding support. All authors have read and approved the final manuscript.

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Data availability
Publicly available datasets were analyzed in this study.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
All authors have agreed on the contents of the manuscript.

Competing interests
All authors declare that they have no competing interests.

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